

Angel Chasing!*

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today's session on chasing angels

-Abhay

Remark. There are close to 50 problems in the handout so it would be fairly unreasonable for you to be able to do all of them and not all may even appeal to your tastes. I have tried to ensure that the problems are nice but you should make the call on what you like and try problems accordingly. The aim should not be to solve all the problems but learn as much from each problem as you can.

I hope you enjoy them! And please don't worry too much if you are unable to solve and just try to learn as much as you can. If you're comfortable with most problems, maybe look at the problem set or the last few problems of any given section. If you've done all of them and want yet more problems, feel free to mail me at rg.meow@outlook.com

§1 Introductory

Note. As this section is only meant as an introductory section to new students, I've not really tried to make the sources more random and I hope being able to solve IMO problems can motivate everyone to try harder problems as well!

Problem 1.1 (ISL 2019 G1). Let ABC be a triangle. Circle Γ passes through A , meets segments AB and AC again at points D and E respectively, and intersects segment BC at F and G such that F lies between B and G . The tangent to circle BDF at F and the tangent to circle CEG at G meet at point T . Suppose that points A and T are distinct. Prove that line AT is parallel to BC .

Problem 1.2 (IMO 2020/1). Consider the convex quadrilateral $ABCD$. The point P is in the interior of $ABCD$. The following ratio equalities hold:

$$\angle PAD : \angle PBA : \angle DPA = 1 : 2 : 3 = \angle CBP : \angle BAP : \angle BPC$$

Prove that the following three lines meet in a point: the internal bisectors of angles $\angle ADP$ and $\angle PCB$ and the perpendicular bisector of segment AB .

Problem 1.3 (INMO 2021/5). In a convex quadrilateral $ABCD$, $\angle ABD = 30^\circ$, $\angle BCA = 75^\circ$, $\angle ACD = 25^\circ$ and $CD = CB$. Extend CB to meet the circumcircle of triangle DAC at E . Prove that $CE = BD$.

*The title is intentional.

Problem 1.4 (USAMO 2021/1). Rectangles BCC_1B_2 , CAA_1C_2 , and ABB_1A_2 are erected outside an acute triangle ABC . Suppose that

$$\angle BC_1C + \angle CA_1A + \angle AB_1B = 180^\circ.$$

Prove that lines B_1C_2 , C_1A_2 , and A_1B_2 are concurrent.

Problem 1.5 (IMO 2018/1). Let Γ be the circumcircle of acute triangle ABC . Points D and E are on segments AB and AC respectively such that $AD = AE$. The perpendicular bisectors of BD and CE intersect minor arcs AB and AC of Γ at points F and G respectively. Prove that lines DE and FG are either parallel or they are the same line.

Problem 1.6 (INMO 2020/1). Let Γ_1 and Γ_2 be two circles of unequal radii, with centres O_1 and O_2 respectively, intersecting in two distinct points A and B . Assume that the centre of each circle is outside the other circle. The tangent to Γ_1 at B intersects Γ_2 again in C , different from B ; the tangent to Γ_2 at B intersects Γ_1 again at D , different from B . The bisectors of $\angle DAB$ and $\angle CAB$ meet Γ_1 and Γ_2 again in X and Y , respectively. Let P and Q be the circumcentres of triangles ACD and XAY , respectively. Prove that PQ is the perpendicular bisector of the line segment O_1O_2 .

Problem 1.7 (IMO 2017/4). Let R and S be different points on a circle Ω such that RS is not a diameter. Let ℓ be the tangent line to Ω at R . Point T is such that S is the midpoint of the line segment RT . Point J is chosen on the shorter arc RS of Ω so that the circumcircle Γ of triangle JST intersects ℓ at two distinct points. Let A be the common point of Γ and ℓ that is closer to R . Line AJ meets Ω again at K . Prove that the line KT is tangent to Γ .

Problem 1.8 (ISL 2011/G1). Let ABC be an acute triangle. Let ω be a circle whose centre L lies on the side BC . Suppose that ω is tangent to AB at B' and AC at C' . Suppose also that the circumcentre O of triangle ABC lies on the shorter arc $B'C'$ of ω . Prove that the circumcircle of ABC and ω meet at two points.

Problem 1.9 (EGMO 2021/4). Let ABC be a triangle with incenter I and let D be an arbitrary point on the side BC . Let the line through D perpendicular to BI intersect CI at E . Let the line through D perpendicular to CI intersect BI at F . Prove that the reflection of A across the line EF lies on the line BC .

Problem 1.10 (IMO 2019/2). In triangle ABC , point A_1 lies on side BC and point B_1 lies on side AC . Let P and Q be points on segments AA_1 and BB_1 , respectively, such that PQ is parallel to AB . Let P_1 be a point on line PB_1 , such that B_1 lies strictly between P and P_1 , and $\angle PP_1C = \angle BAC$. Similarly, let Q_1 be the point on line QA_1 , such that A_1 lies strictly between Q and Q_1 , and $\angle CQ_1Q = \angle CBA$.

Prove that points P, Q, P_1 , and Q_1 are concyclic.

§2 Radical Axis and Power of Point

Handout to refer if you have not heard the terms before: [Yufei Zhao's PoP](#)

Lemma 2.1 (Euler's relation)

Let ABC be a triangle with circumcenter O and incenter I and R circumradius and r in radius. Prove that $OI^2 = R(R - 2r)$.

Problem 2.2 (APMO 2020/1). Let Γ be the circumcircle of $\triangle ABC$. Let D be a point on the side BC . The tangent to Γ at A intersects the parallel line to BA through D at point E . The segment CE intersects Γ again at F . Suppose B, D, F, E are concyclic. Prove that AC, BF, DE are concurrent.

Problem 2.3 (IMO 1995/1). Let A, B, C, D be four distinct points on a line, in that order. The circles with diameters AC and BD intersect at X and Y . The line XY meets BC at Z . Let P be a point on the line XY other than Z . The line CP intersects the circle with diameter AC at C and M , and the line BP intersects the circle with diameter BD at B and N . Prove that the lines AM, DN, XY are concurrent.

Problem 2.4 (IMO 2008/1). Let H be the orthocenter of an acute-angled triangle ABC . The circle Γ_A centered at the midpoint of BC and passing through H intersects the sideline BC at points A_1 and A_2 . Similarly, define the points B_1, B_2, C_1 and C_2 .

Prove that the six points A_1, A_2, B_1, B_2, C_1 and C_2 are concyclic.

Problem 2.5 (IMO 2000/1). Two circles G_1 and G_2 intersect at two points M and N . Let AB be the line tangent to these circles at A and B , respectively, so that M lies closer to AB than N . Let CD be the line parallel to AB and passing through the point M , with C on G_1 and D on G_2 . Lines AC and BD meet at E ; lines AN and CD meet at P ; lines BN and CD meet at Q . Show that $EP = EQ$.

Problem 2.6 (USAMO 1998/2). Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be concentric circles, with $\mathcal{C}_2$ in the interior of $\mathcal{C}_1$. From a point A on $\mathcal{C}_1$ one draws the tangent AB to $\mathcal{C}_2$ ($B \in \mathcal{C}_2$). Let C be the second point of intersection of AB and $\mathcal{C}_1$, and let D be the midpoint of AB . A line passing through A intersects $\mathcal{C}_2$ at E and F in such a way that the perpendicular bisectors of DE and CF intersect at a point M on AB . Find, with proof, the ratio AM/MC .

Problem 2.7 (ISL 2011 G2). Let $A_1A_2A_3A_4$ be a non-cyclic quadrilateral. Let O_1 and r_1 be the circumcentre and the circumradius of the triangle $A_2A_3A_4$. Define O_2, O_3, O_4 and r_2, r_3, r_4 in a similar way. Prove that

$$\frac{1}{O_1A_1^2 - r_1^2} + \frac{1}{O_2A_2^2 - r_2^2} + \frac{1}{O_3A_3^2 - r_3^2} + \frac{1}{O_4A_4^2 - r_4^2} = 0.$$

§3 Simson Line and Steiner Point

Lemma 3.1 (Simson Line)

Let $ABCD$ be a quadrilateral. Let D_A, D_B, D_C be feet of perpendicular from D to BC, AC, AB . Prove that D_A, D_B, D_C are collinear iff $ABCD$ is cyclic.

Lemma 3.2 (Simson Line Bisection)

Let $ABCD$ be cyclic and H be orthocenter of ABC . The simson line formed by feet of perpendiculars of D bisects line DH .

Corollary 3.3 (Steiner Point/Steiner Line)

Let ABC be a triangle with orthocenter H and ℓ be a line passing through H . Reflect ℓ over BC, AC, AB to get ℓ_A, ℓ_B, ℓ_C then ℓ_A, ℓ_B, ℓ_C concur on (ABC) .

Problem 3.4 (Unknown). Let ω_1, ω_2 be two circles with centers O_1, O_2 . Let S be the set of lines which cut equal length chords in both circles. Let $l_1, l_2, l_3 \in S$. Prove that midpoint of O_1, O_2 lies on the circumcircle of the triangle formed by l_1, l_2, l_3 .

Problem 3.5 (ISL 2018 G4). A point T is chosen inside a triangle ABC . Let A_1, B_1 , and C_1 be the reflections of T in BC, CA , and AB , respectively. Let Ω be the circumcircle of the triangle $A_1B_1C_1$. The lines A_1T, B_1T , and C_1T meet Ω again at A_2, B_2 , and C_2 , respectively. Prove that the lines AA_2, BB_2 , and CC_2 are concurrent on Ω .

Problem 3.6 (IMO 2007/2). Consider five points A, B, C, D and E such that $ABCD$ is a parallelogram and $BCED$ is a cyclic quadrilateral. Let ℓ be a line passing through A . Suppose that ℓ intersects the interior of the segment DC at F and intersects line BC at G . Suppose also that $EF = EG = EC$. Prove that ℓ is the bisector of angle DAB .

Problem 3.7 (CAMO 2020/4). Let ABC be a triangle and Q a point on its circumcircle. Let E and F be the reflections of Q over $\overline{AB}$ and $\overline{AC}$, respectively. Select points X and Y on line EF such that $\overline{BX} \parallel \overline{AC}$ and $\overline{CY} \parallel \overline{AB}$, and let M and N be the reflections of X and Y over B and C respectively. Prove that M, Q, N are collinear.

Problem 3.8 (USATSTST 2018/5). Let ABC be an acute triangle with circumcircle ω , and let H be the foot of the altitude from A to $\overline{BC}$. Let P and Q be the points on ω with $PA = PH$ and $QA = QH$. The tangent to ω at P intersects lines AC and AB at E_1 and F_1 respectively; the tangent to ω at Q intersects lines AC and AB at E_2 and F_2 respectively. Show that the circumcircles of $\triangle AE_1F_1$ and $\triangle AE_2F_2$ are congruent, and the line through their centers is parallel to the tangent to ω at A .

Problem 3.9 (IMO 2011/6). Let ABC be an acute triangle with circumcircle Γ . Let ℓ be a tangent line to Γ , and let ℓ_a, ℓ_b and ℓ_c be the lines obtained by reflecting ℓ in the lines BC, CA and AB , respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b and ℓ_c is tangent to the circle Γ .

§4 Problem Set

Note. The problem quality in this section is fairly high but I've been careful not to put in any absolutely ridiculous problem and all of the problems here will be very solvable.

Problem 4.1 (Classic). The orthocenter, circumcenter, incenter and one vertex of an acute scalene triangle are concyclic. Prove that one of the angles of the triangle is $\frac{\pi}{3}$.

Problem 4.2 (Iran 2020/3/G/2). Triangle ABC with its circumcircle Γ is given. Points D and E are chosen on segment BC such that $\angle BAD = \angle CAE$. The circle ω is tangent to AD at A with its circumcenter lies on Γ . Reflection of A through BC is A' . If the line $A'E$ meet ω at L and K . Then prove either BL and CK or BK and CL meet on Γ .

Problem 4.3 (Poland 2019). Let X be a point lying in the interior of the acute triangle ABC such that

$$\angle BAX = 2\angle XBA \quad \angle XAC = 2\angle XCA$$

Denote by M the midpoint of the arc BAC . Prove that $XM = XA$.

Problem 4.4 (Canada Buffet Contest 2021 G4). In $\triangle ABC$, let D, E, F be the feet of altitudes from A, B, C respectively. Let K be the orthocenter of $\triangle DEF$. Let U, V, W be the projections of points F, D, E , on BC, CA, AB respectively. Assume that the circumcircles of $\triangle ABC$ and $\triangle UVW$ intersect at two points M and N , prove that $KM = KN$.

Problem 4.5 (Iran TST 2019/4). Consider triangle ABC with orthocenter H . Let points M and N be the midpoints of segments BC and AH . Point D lies on line MH so that $AD \parallel BC$ and point K lies on line AH so that $DNMK$ is cyclic. Points E and F lie on lines AC and AB such that $\angle EHM = \angle C$ and $\angle FHM = \angle B$. Prove that points D, E, F and K lie on a circle.

Problem 4.6 (ISL 2011 G4). Let ABC be an acute triangle with circumcircle Ω . Let B_0 be the midpoint of AC and let C_0 be the midpoint of AB . Let D be the foot of the altitude from A and let G be the centroid of the triangle ABC . Let ω be a circle through B_0 and C_0 that is tangent to the circle Ω at a point $X \neq A$. Prove that the points D, G and X are collinear.

Problem 4.7 (Iran TST 2010/5). Circles W_1, W_2 intersect at P, K . XY is common tangent of two circles which is nearer to P and X is on W_1 and Y is on W_2 . XP intersects W_2 for the second time in C and YP intersects W_1 in B . Let A be intersection point of BX and CY . Prove that if Q is the second intersection point of circumcircles of ABC and AXY

$$\angle QXA = \angle QKP$$

Problem 4.8 (IGO 2018/5). $ABCD$ is a cyclic quadrilateral. A circle passing through A, B is tangent to segment CD at point E . Another circle passing through C, D is tangent to AB at point F . Point G is the intersection point of AE, DF , and point H is the intersection point of BE, CF . Prove that the incenters of triangles AGF, BHF, CHE, DGE lie on a circle.

Problem 4.9 (RMMSL 2016/G1). Two circles, ω_1 and ω_2 , centred at O_1 and O_2 , respectively, meet at points A and B . A line through B meets ω_1 again at C , and ω_2 again at D . The tangents to ω_1 and ω_2 at C and D , respectively, meet at E , and the line AE meets the circle ω through A, O_1, O_2 again at F . Prove that the length of the segment EF is equal to the diameter of ω .

Problem 4.10 (EGMO 2021/3). Let ABC be a triangle with an obtuse angle at A . Let E and F be the intersections of the external bisector of angle A with the altitudes of ABC through B and C respectively. Let M and N be the points on the segments EC and FB respectively such that $\angle EMA = \angle BCA$ and $\angle ANF = \angle ABC$. Prove that the points E, F, N, M lie on a circle.

Problem 4.11 (Iran TST 2017/18). In triangle ABC let O and H be the circumcenter and the orthocenter. The point P is the reflection of A with respect to OH . Assume that P is not on the same side of BC as A . Points E, F lie on AB, AC respectively such that $BE = PC$, $CF = PB$. Let K be the intersection point of AP, OH . Prove that $\angle EKF = 90^\circ$.

Problem 4.12 (ISL 2011 G6). Let ABC be a triangle with $AB = AC$ and let D be the midpoint of AC . The angle bisector of $\angle BAC$ intersects the circle through D, B and C at the point E inside the triangle ABC . The line BD intersects the circle through A, E and B in two points B and F . The lines AF and BE meet at a point I , and the lines CI and BD meet at a point K . Show that I is the incentre of triangle KAB .

Problem 4.13 (USATST 2016/2). Let ABC be a scalene triangle with circumcircle Ω , and suppose the incircle of ABC touches BC at D . The angle bisector of $\angle A$ meets BC and Ω at E and F . The circumcircle of $\triangle DEF$ intersects the A -excircle at S_1, S_2 , and Ω at $T \neq F$. Prove that line AT passes through either S_1 or S_2 .

Problem 4.14 (Taiwan TST Round 3 2021). Let ABC be a triangle with $AB < AC$, and let I_a be its A -excenter. Let D be the projection of I_a to BC . Let X be the intersection of AI_a and BC , and let Y, Z be the points on AC, AB , respectively, such that X, Y, Z are on a line perpendicular to AI_a . Let the circumcircle of AYZ intersect AI_a again at U . Suppose that the tangent of the circumcircle of ABC at A intersects BC at T , and the segment TU intersects the circumcircle of ABC at V . Show that $\angle BAV = \angle DAC$.

Problem 4.15 (IMEO 2019/6). Let ABC be a scalene triangle with incenter I and circumcircle ω . The internal and external bisectors of angle $\angle BAC$ intersect BC at D and E , respectively. Let M be the point on segment AC such that $MC = MB$. The tangent to ω at B meets MD at S . The circumcircles of triangles ADE and BIC intersect each other at P and Q . If AS meets ω at a point K other than A , prove that K lies on PQ .

Problem 4.16 (RMMSL 2018/G1). Let ABC be a triangle and let H be the orthogonal projection of A on the line BC . Let K be a point on the segment AH such that $AH = 3KH$. Let O be the circumcenter of triangle ABC and let M and N be the midpoints of sides AC and AB respectively. The lines KO and MN meet at a point Z and the perpendicular at Z to OK meets lines AB, AC at X and Y respectively. Show that $\angle XKY = \angle CKB$.

Problem 4.17 (Taiwan TST Round 2 2021). Let ABC be a scalene triangle, and points O and H be its circumcenter and orthocenter, respectively. Point P lies inside triangle AHO and satisfies $\angle AHP = \angle POA$. Let M be the midpoint of segment $\overline{OP}$. Suppose that BM and CM intersect with the circumcircle of triangle ABC again at X and Y , respectively.

Prove that line XY passes through the circumcenter of triangle APO .

Problem 4.18 (Infinity Dots MO 2017/3). Let $\triangle ABC$ be an acute triangle with circumcenter O and orthocenter H . The line through O parallel to BC intersect AB at D and AC at E . X is the midpoint of AH . Prove that the circumcircles of $\triangle BD X$ and $\triangle CE X$ intersect again at a point on line AO .

Problem 4.19 (CAMO 2020/3). Let ABC be a triangle with incircle ω , and let ω touch $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ at D , E , F , respectively. Point M is the midpoint of $\overline{EF}$, and T is the point on ω such that $\overline{DT}$ is a diameter. Line MT meets the line through A parallel to $\overline{BC}$ at P and ω again at Q . Lines DF and DE intersect line AP at X and Y respectively. Prove that the circumcircles of $\triangle APQ$ and $\triangle DXY$ are tangent.

Problem 4.20 (ISL 2017 G7). A convex quadrilateral $ABCD$ has an inscribed circle with center I . Let I_a, I_b, I_c and I_d be the incenters of the triangles DAB, ABC, BCD and CDA , respectively. Suppose that the common external tangents of the circles AI_bI_d and CI_bI_d meet at X , and the common external tangents of the circles BI_aI_c and DI_aI_c meet at Y . Prove that $\angle XIY = 90^\circ$.

Problem 4.21 (Taiwan TST 2021 Round 1). Let ABC be a triangle with incenter I and circumcircle Ω . A point X on Ω which is different from A satisfies $AI = XI$. The incircle touches AC and AB at E, F , respectively. Let M_a, M_b, M_c be the midpoints of sides BC, CA, AB , respectively. Let T be the intersection of the lines M_bF and M_cE . Suppose that AT intersects Ω again at a point S .

Prove that X, M_a, S, T are concyclic.

Problem 4.22 (LMAO 2021/4). Let P, Q be chosen inside triangle ABC such that $\angle PBC = \angle ABQ$ and $\angle PCB = \angle ACQ$. The tangent to (BPC) at P and the tangent to (BQC) at Q intersect at R . Prove that circles (PQR) and (ABC) , and line AR share a common point.

Problem 4.23 (Iran TST 2011/6). The circle ω with center O has given. From an arbitrary point T outside of ω draw tangents TB and TC to it. K and H are on TB and TC respectively.

a) B' and C' are the second intersection point of OB and OC with ω respectively. K' and H' are on angle bisectors of $\angle BCO$ and $\angle CBO$ respectively such that $KK' \perp BC$ and $HH' \perp BC$. Prove that K, H', B' are collinear if and only if H, K', C' are collinear.

b) Consider there exist two circle in TBC such that they are tangent two each other at J and both of them are tangent to ω . and one of them is tangent to TB at K and other one is tangent to TC at H . Prove that two quadrilateral $BKJI$ and $CHJI$ are cyclic (I is incenter of triangle OBC).

Problem 4.24 (IGO 2019/4). Given an acute non-isosceles triangle ABC with circum-circle Γ . M is the midpoint of segment BC and N is the midpoint of arc BC of Γ (the one that doesn't contain A). X and Y are points on Γ such that $BX \parallel CY \parallel AM$. Assume there exists point Z on segment BC such that circumcircle of triangle XYZ is tangent to BC . Let ω be the circumcircle of triangle ZMN . Line AM meets ω for the second time at P . Let K be a point on ω such that $KN \parallel AM$, ω_b be a circle that passes through B, X and tangents to BC and ω_c be a circle that passes through C, Y and tangents to BC . Prove that circle with center K and radius KP is tangent to 3 circles ω_b, ω_c and Γ .

Problem 4.25 (Russia 2021/11.4). In triangle ABC angle bisectors AA_1 and CC_1 intersect at I . Line through B parallel to AC intersects rays AA_1 and CC_1 at points A_2 and C_2 respectively. Let O_a and O_c be the circumcenters of triangles AC_1C_2 and CA_1A_2 respectively. Prove that $\angle O_aBO_c = \angle AIC$

Problem 4.26 (IMO 2018/6). A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. Point X lies inside $ABCD$ so that

$$\angle XAB = \angle XCD \quad \text{and} \quad \angle XBC = \angle XDA.$$

Prove that $\angle BXA + \angle DXC = 180^\circ$.

Problem 4.27 (Vietnam TST 2021/3). Let ABC be a triangle and N be a point that differs from A, B, C . Let A_b be the reflection of A through NB , and B_a be the reflection of B through NA . Similarly, we define B_c, C_b, A_c, C_a . Let m_a be the line through N and perpendicular to B_cC_b . Define similarly m_b, m_c .

- a) Assume that N is the orthocenter of $\triangle ABC$, show that the respective reflection of m_a, m_b, m_c through the bisector of angles $\angle BNC, \angle CNA, \angle ANB$ are the same line.
- b) Assume that N is the nine-point center of $\triangle ABC$, show that the respective reflection of m_a, m_b, m_c through BC, CA, AB concur.

Problem 4.28 (IMO 2019/6). Let I be the incentre of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC, CA , and AB at D, E , and F , respectively. The line through D perpendicular to EF meets ω at R . Line AR meets ω again at P . The circumcircles of triangle PCE and PBF meet again at Q .

Prove that lines DI and PQ meet on the line through A perpendicular to AI .

Problem 4.29 (ISL 2018 G7). Let O be the circumcentre, and Ω be the circumcircle of an acute-angled triangle ABC . Let P be an arbitrary point on Ω , distinct from A, B, C , and their antipodes in Ω . Denote the circumcentres of the triangles AOP, BOP , and COP by O_A, O_B , and O_C , respectively. The lines ℓ_A, ℓ_B, ℓ_C perpendicular to BC, CA , and AB pass through O_A, O_B , and O_C , respectively. Prove that the circumcircle of triangle formed by ℓ_A, ℓ_B , and ℓ_C is tangent to the line OP .

§5 Other Nice Handouts

I will add a few nice places to refer to for good geometry problems and handouts in general-

1. [Yufei Zhao's handouts](#)
2. [Canada IMO training camp](#)
3. [Euclidian Geometry in Maths Olympiad by Evan Chen](#)
4. [Handouts by Evan Chen](#)
5. [A Beautiful Journey Through Olympiad Geometry](#)
6. [Eric Shen's handouts](#)